

University of Dhaka

Department of Computer Science and Engineering

Project Report: Object Oriented Programming (CSE-2102)

Project Name: ATM Simulator System

Team Members

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Introduction

In today's rapidly evolving financial landscape, Automated Teller Machines (ATMs) play a crucial role in providing convenient and efficient banking services to millions of people worldwide. The "ATM Simulator System" project aims to emulate the functionality of a real-world ATM, offering users a virtual environment to perform typical banking transactions such as cash withdrawals, deposits, balance inquiries, and fund transfers. This simulation is designed to demonstrate the key components and processes involved in ATM operations, highlighting the integration of hardware and software to deliver a seamless user experience.

The primary objective of this project is to create a comprehensive and interactive ATM simulator that can serve as an educational tool for understanding the inner workings of ATMs. By replicating the user interface and transaction workflows of a physical ATM, the simulator provides an immersive learning experience for students, developers, and banking professionals. It allows users to gain insights into the security protocols, transaction processing, and error handling mechanisms that are essential for the reliable operation of ATMs.

This project report outlines the development process of the ATM simulator, detailing the system architecture, design considerations, and implementation techniques. It covers the following key aspects:

The ATM simulator system serves as a valuable tool for understanding the complexities of ATM operations and the importance of secure and efficient transaction processing in the banking sector. By providing a realistic

simulation environment, this project not only aids in educational and training purposes but also paves the way for future innovations in ATM technology.

Objectives

The primary objective of the "ATM Simulator System" project is to develop a comprehensive, interactive simulation of an Automated Teller Machine (ATM) that replicates the functionalities and user experiences of real-world ATMs. This project aims to achieve the following specific goals:

- 1. Educational Tool:** To create an educational platform that provides users with a detailed understanding of ATM operations, including the technical and security aspects involved in banking transactions.
- 2. User Interface Simulation:** To design a user-friendly interface that mimics the look and feel of an actual ATM, enabling users to perform typical transactions such as withdrawals, deposits, balance inquiries, and fund transfers.
- 3. Transaction Workflow Emulation:** To accurately simulate the transaction processing workflows, including data validation, transaction authorization, and error handling, reflecting real-world banking scenarios.
- 4. Security Implementation:** To integrate robust security measures that protect user data and ensure the integrity of transactions, demonstrating best practices in financial security.
- 5. System Architecture Understanding:** To provide insights into the system architecture of ATMs, including hardware components and software interactions, thereby enhancing users' technical knowledge.
- 6. Practical Application:** To offer a practical application for students and professionals to test and understand the principles of ATM technology, including software development, network communication, and database management.
- 7. Testing and Validation:** To implement thorough testing and validation processes to ensure the simulator functions correctly and reliably under

various scenarios, demonstrating the importance of quality assurance in software development.

8. Foundation for Future Enhancements: To establish a scalable and modular framework that can be easily extended with additional features and improvements, encouraging ongoing innovation in ATM technology.

By achieving these objectives, the ATM Simulator System aims to serve as a valuable resource for learning, development, and research, fostering a deeper appreciation of the technologies that drive modern banking services.

Project Features and Modules

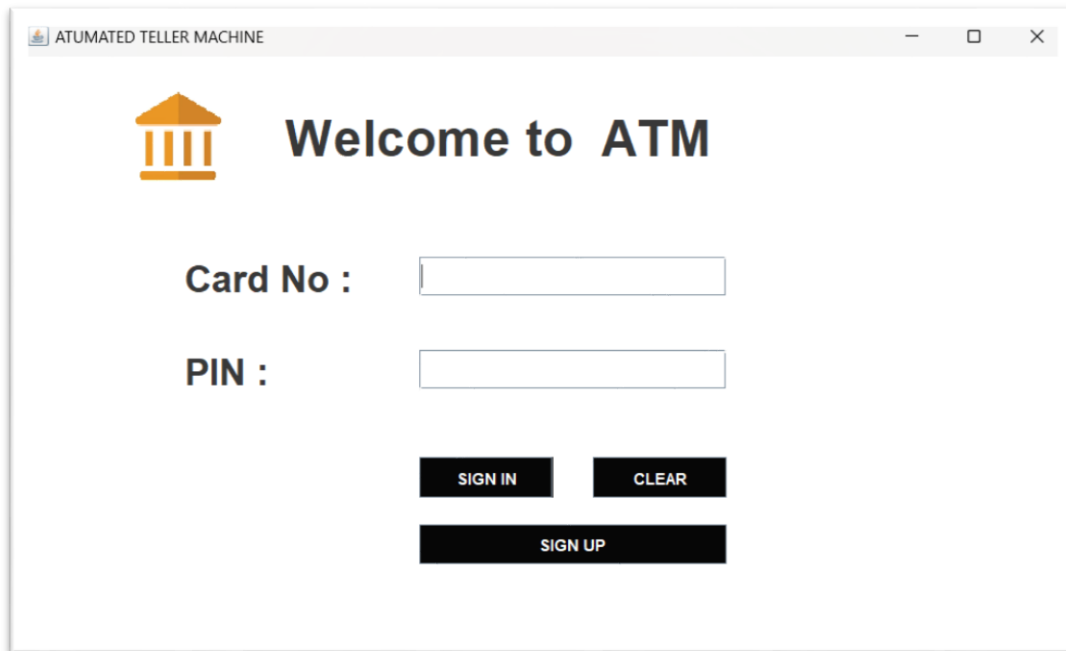
Secure ATM Login Interface Implementation

1. User Interface Design: The login window is designed using Swing components such as `JLabel`, `TextField`, `PasswordField`, and `Button`. The interface includes fields for card number and PIN, as well as buttons for login, clearing input fields, and signing up for a new account.

2. Event Handling: The class implements `ActionListener` to handle button click events. Depending on the button clicked, the appropriate action is taken, such as clearing input fields, validating login credentials, or navigating to the sign-up interface.

3.. Database Interaction: The code interacts with a database to validate user credentials using a SQL query. This is facilitated by the `Conn` class (presumably a custom class for database connections) and involves executing a query to check if the entered card number and PIN match any records.

This login interface is a crucial component of the ATM simulator, providing users with a secure entry point to perform banking transactions.



User Registration Interface for ATM Simulator System

This code implements a user registration interface for an ATM simulator system using Java Swing components. The primary features of this implementation include collecting personal details, handling user input, and transitioning to the next registration phase. Here is a detailed description of the key components.

1. **Form Generation:** The interface generates a unique application form number using a random number generator. This form number is displayed prominently at the top of the interface.
2. **User Interface Design:** The registration form is designed with various `JLabel` and `JTextField` components to collect user information such as name, father's name, date of birth, gender, email, marital status, address, city, district, and pin code. The form is visually organized for easy data entry.

APPLICATION FORM NO. 5653

Page 1 : Personal Details

Name :

Father's Name :

Date of Birth :

Gender : ☐ male ☐ female

Email Address :

Marital Status: ☐ married ☐ unmarried ☐ others

Address :

District :

Division :

Pin Code :

NEXT

3. Radio Buttons for Selection: Gender and marital status fields utilize `JRadioButton` components grouped with `ButtonGroup` to ensure only one option can be selected at a time, preventing invalid input.

4. Data Validation and Storage: Before storing user data in the database, the input is validated for completeness. Missing fields trigger a `JOptionPane` error message. If all required fields are filled, the data is inserted into the database using a SQL `INSERT` statement executed via the `Conn` class.

Additional User Details Interface for ATM Simulator System

1. Form Structure and Layout: The interface continues from the previous registration step by asking for more detailed personal and demographic

information. It displays the form number passed from the previous step and labels for each field.

2. User Interface Design: The form uses `JLabel`, `JTextField`, `JRadioButton`, and `JComboBox` components to collect information such as religion, category, income, educational qualification, occupation, phone number, and NID

Page 2 : Additional Details

Religion:

Category:

Income:

Educational

Qualification:

Occupation:

Phone Number:

NID Number:

Bangladeshi Citizen: ☐ Yes ☐ No

Existing Account: ☐ Yes ☐ No

NEXT

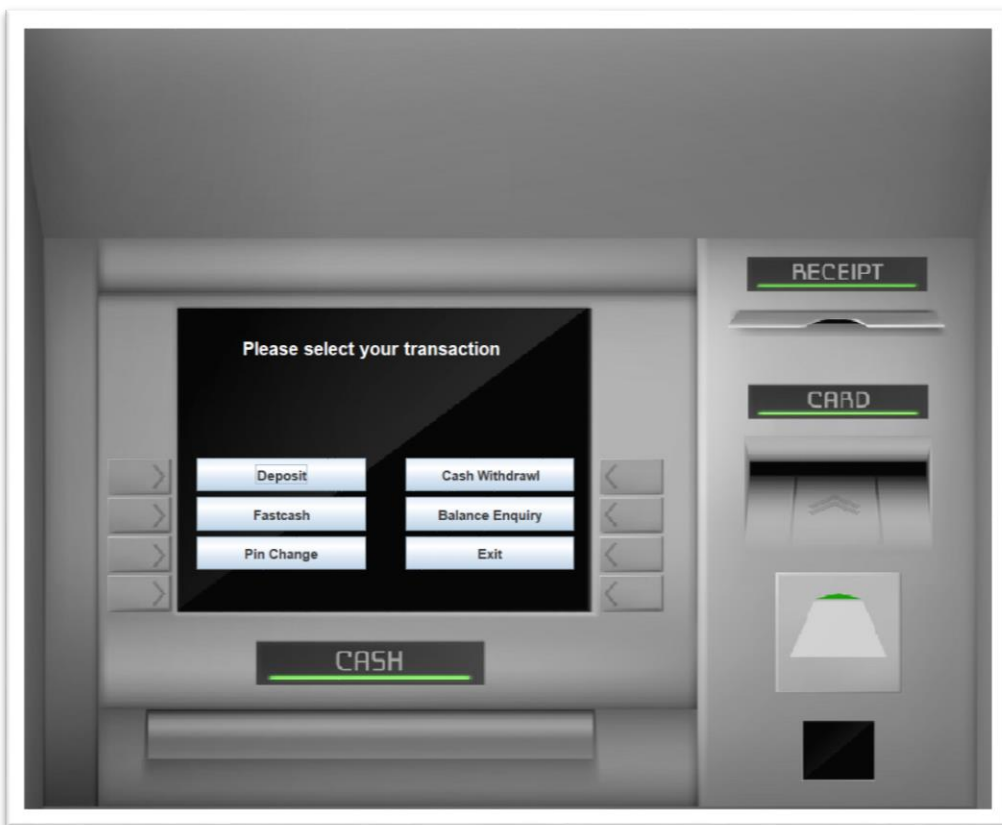
number. The layout is structured to enhance user interaction and ease of data entry.

3. ComboBox for Selections: Dropdown menus (`JComboBox`) are used for fields with predefined options such as religion, category, income, educational qualification, and occupation, ensuring consistent data input.

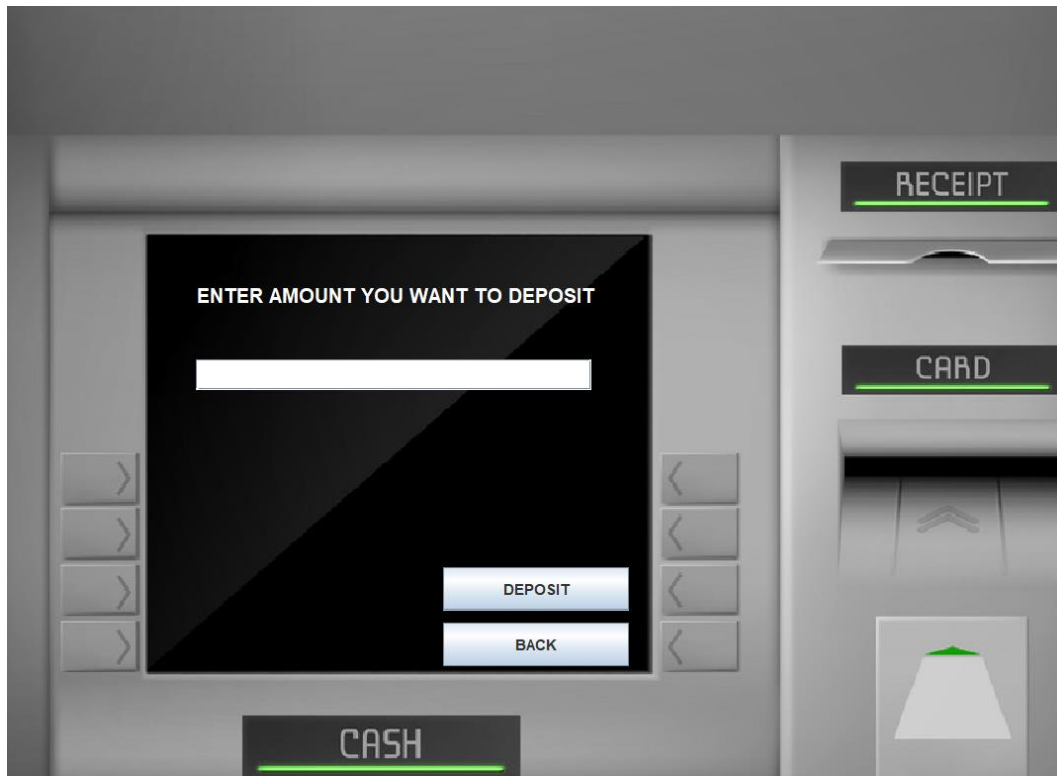
4. Radio Buttons for Binary Choices: The form includes `JRadioButton` components for yes/no questions regarding Bangladeshi citizenship and existing account status. These are grouped using `ButtonGroup` to ensure mutual exclusivity.

5. Action Handling: The class implements `ActionListener` to manage the 'Next' button click event. Upon clicking, it validates the input fields, ensuring all required information is provided. If validation succeeds, it saves the data to the database and proceeds to the next registration step (`SignupThree`).

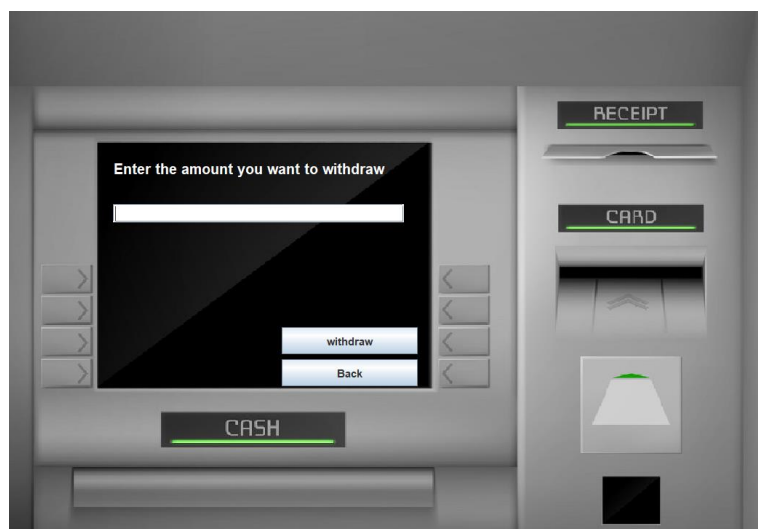
ATM Transaction Interface for ATM Simulator System



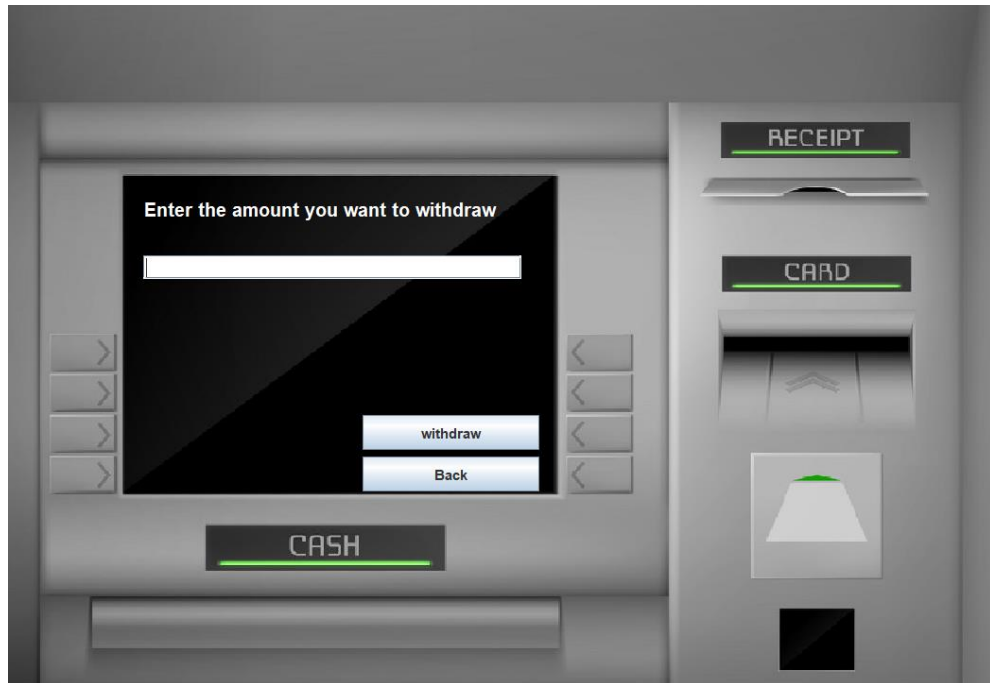
ATM Deposit Interface Implementation in Java



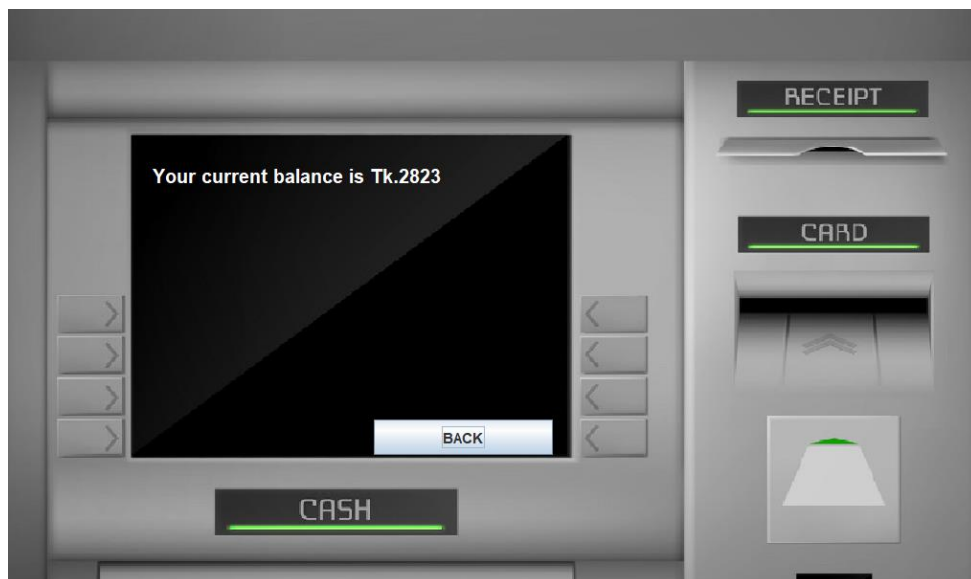
ATM Withdrawl Interface Implementation in Java



ATM Pinchange Interface Implementation in Java



ATM BalaceEnquiry Interface Implementation in Java



Team Member Responsibilities:

In the collaborative development of "ATM simulator system" each team member has taken on specific responsibilities, contributing to different modules of the project. The distribution of responsibilities is as follows:

Name :Tawyabul Islam Tamim

Responsibility: Database management system, Transactions class, FastCash class, Conn class, deposit class,Withdrawl class,pinchange class,balanceEnquiry class and UI design

Name: Ovijit Chandra Balo

Responsibility: SignupTwo class,SignupThree class ,Database management system

Name: Abdullah Al Arman Emon

Responsibility: login class, Signup One class, Overall design planning, UI design, Code debugging

- Programming Languages: JAVA
 1. Library: JAVA library
 - **java.awt**
 - **java.awt.event**
 - **javax.swing**
 - **java.util**
 2. GitHub: Used for project collaboration
 3. IDE (Integrated Development Environment): **NetBeans** and Visual Studio Code
 - Visual Studio Code: Offers a powerful, customizable, and feature-rich IDE suitable for collaborative development.
 - Yes, NetBeans is an Integrated Development Environment (IDE). It is used primarily for developing Java applications, but it also supports other

programming languages such as PHP, C/C++, HTML5, and JavaScript. NetBeans provides features like code editing, debugging, and project management, making it easier for developers to write, test, and maintain their code.

Conclusion

The ATM Deposit Interface project successfully demonstrates the creation of a simple and user-friendly GUI application for depositing money in an ATM simulator. By utilizing Java's Swing library for the GUI components and AWT for event handling, the project provides a practical example of integrating front-end user interactions with backend database operations. The implementation ensures that user inputs are processed and stored efficiently, with appropriate feedback provided to the user.

Key Learnings

1. GUI Design with Swing and AWT:

The project highlights the use of Java Swing and AWT libraries to design and manage a graphical user interface. This includes creating and positioning components like labels, text fields, and buttons, and handling user actions using event listeners.

2. Database Integration: A critical aspect of this project is the interaction with a backend database to record deposit transactions. It demonstrates how to execute SQL queries from a Java application, handle database connections, and ensure data is accurately stored and retrieved. This provides a foundation for understanding how frontend applications can communicate with databases in real-world scenarios.

References:

<https://lazyfoo.net/tutorials/SDL/>

<https://spritedatabase.net/game/849>

